Dear Editor,

Greetings of the day!

We would like to submit our manuscript entitled "**SecChaos-RL: A Threat-Aware Self-Adaptive Resilience Engineering Framework for Secure Software Systems**" for publication in the "**Software Quality Journal"**. The manuscript is original research that has not been published before and not submitted to any other journal. The manuscript has been read and approved by all the authors.

In today's microservices-driven software applications, security and resilience are growing concerns with the complexity of interdependency relationships and the variety of emerging threats. Current Security Chaos Engineering techniques are primarily static and manual, and are unable to offer autonomous and adaptive mitigation.

To overcome this limitation, we present SecChaos-RL, an autonomous threat-aware mitigation framework based on reinforcement learning techniques with Security Chaos Engineering. The framework is modeled as a multi-objective Markov Decision Process, includes 8 security fault categories, threat-aware representation learning and adaptive mitigation strategies.

SecChaos-RL has been evaluated in an experiment across 30 independent random seeds, and achieves 83.45% risk reduction, 96.26% service availability, 0.17 steps MTTR, 4.91% attack success rate, and an 88.95% resilience score, all outperforming the DQN and A2C and PPO baselines.

We felt this work fits well with the "**Software Quality Journal"** due to its focus on software security, software resilience, software self-adaptation, and quantitative software quality evaluation.

We would like to thank you for your consideration of our manuscript. We appreciate your time and look forward to your response.

Yours sincerely,

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